Predicting When Cheap Simulators Break Neural Inference of Bursty Gene Expression

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Abstract

Amortized simulation-based inference (SBI) often trains its neural posterior estimator on a cheap approximate simulator rather than on the exact model that generated the data, and the resulting misspecification can silently break posterior calibration. We study this failure in stochastic gene expression in single cells, a setting where the exact answer is known and where practitioners routinely replace the exact chemical master equation (CME), a discrete Markov model of molecule counts, with a cheaper continuous approximation such as the chemical Langevin equation (CLE). We ask how this substitution biases inference, and whether we can predict the resulting failure pre-inference rather than merely detect it after the fact. Using the two-state (telegraph) model, whose exact stationary distribution and moments have closed form, we show that the CLE misspecifies the model directionally in the bursty, low-copy regime, where it overestimates mean expression, suppresses the Fano factor, and erases the zero-inflation that signals bursty transcription. We train neural posterior estimators on the cheap simulators and on the exact CME and evaluate all against exact CME data. The CLE-trained estimator is biased and overconfident across the whole burstiness range, and a second, structurally different Gaussian surrogate, the linear noise approximation (LNA), is intermediate, less catastrophic than the CLE but with a calibration that collapses once burstiness passes a threshold. An ablation that restricts inference to low-order moments does not rescue it, because enforcing non-negative counts distorts even those moments where bursts dominate. A matched non-Gaussian surrogate that carries the same moments is instead calibrated, which implicates the Gaussian distributional family rather than the moments the surrogate carries. The CME-trained control stays calibrated throughout. Because the telegraph model admits closed-form moments, we anticipate the onset of this failure from the analytic Fano factor, which converts a post hoc diagnosis into a simple per-query routing rule for when a practitioner can trust the cheap simulator. We frame the work as a controlled-sandbox characterization rather than a new inference method; its value lies in a transferable diagnostic for surrogate-simulator misspecification in SBI. Code is available at `https://github.com/Rafther0112/Genetic_SBI`.

1 Introduction

We study the reliability of amortized neural inference when the training simulator only approximates the process that generated the data, using single-cell gene expression as a testbed where the exact model is known. Simulation-based inference (SBI) infers the parameters of a stochastic model from data when the likelihood is intractable but the model can still generate samples (1). Modern SBI trains a neural density estimator, a normalizing flow or a related deep generative model, to approximate the posterior $p(\theta \mid x)$ from pairs (θ, x) that the prior and a forward simulator generate (2; 3). This makes it a natural tool for stochastic gene expression, where single-cell snapshot data (smFISH, scRNA-seq)

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constrain the kinetic parameters of models whose likelihood follows the chemical master equation (CME) (4; 16; 17).

The CME is exact but expensive, because its state space grows combinatorially with the number of species and molecules (4; 15), and an SBI loop must simulate it afresh for every parameter draw. Practitioners therefore substitute cheaper approximate simulators, most commonly the chemical Langevin equation (CLE), a continuous diffusion approximation (5), or its linearization, the linear noise approximation (19; 20). We know these approximations fail precisely in the regime that defines transcription, namely low molecule numbers and bursty, zero-inflated expression (5; 18; 15). In short, the object we can afford to simulate is not the object that generated the data.

Misspecification degrades neural SBI, and a growing body of work tries to detect or robustify against it in general (6; 7; 8). Our setting differs in a way we exploit, because the telegraph model of two-state transcription has a closed-form stationary distribution and closed-form moments (9; 10). We therefore not only measure how the cheap simulator corrupts inference, but also predict where in parameter space it will do so before running any inference. We make four contributions:

- We characterize the CLE's misspecification against the exact CME as *directional and regime-localized*: mean overestimation, Fano suppression, and loss of zero-inflation that all intensify as transcriptional burstiness grows (Section 5.1).
- We show that two Gaussian surrogates degrade the calibration of a neural posterior estimator in distinct ways: the CLE is biased and overconfident across the whole range, while the moment-matched LNA degrades specifically as burstiness grows; a CME-trained control on the same exact CME data escapes both (Section 5.2).
- We narrow down the cause with two controls: a moments-only ablation does not rescue the surrogates, and a non-Gaussian surrogate carrying the *same* moments as the LNA is calibrated, which implicates the Gaussian distributional family rather than cheapness or the moments (Section 5.3).
- We turn the closed-form Fano factor into a pre-inference diagnostic that predicts the onset of failure, which yields a per-query routing rule for when to trust the cheap simulator and when to fall back on the exact one (Section 5.4).

We frame the work as an analysis and diagnostic in a controlled sandbox, not a new inference algorithm. The telegraph model is a testbed, not the deployment target. Here one could sample the Poisson-Beta directly, and we exploit exactly that to validate the diagnostic against a known answer. The setting it stands in for is the common one where no such closed form exists, multi-state promoters, feedback, or coupled genes, and cheap approximate simulators like the CLE and LNA are the only tractable option; there the same diagnostic must run on its empirical, rather than analytic, form. The lesson extends beyond biology to amortized inference in general. An estimator inherits the distributional errors of the simulator that trains it, standard calibration diagnostics reveal those errors only after inference, and a cheap summary of each observation can flag in advance which queries are at risk.

2 Related work

Simulation-based inference. SBI infers parameters of simulators with intractable likelihoods by learning from simulated pairs (1). Neural posterior estimation fits a conditional density estimator, typically a normalizing flow, directly to the posterior (2; 3; 14), and mature tooling has made it routine (11). Our estimator is a standard flow-based NPE; we treat it as fixed machinery and study how its reliability depends on the simulator that trains it.

Misspecification and calibration. A growing literature shows that neural SBI degrades under simulator mismatch and proposes ways to detect or robustify against it (6; 7; 8), while calibration diagnostics such as simulation-based calibration (12) and expected coverage (13) check posterior faithfulness. These lines flag or absorb misspecification after seeing an observation. We instead work where the exact model has a closed form, which lets us predict, before running inference and from a quantity read off the data, where a cheap simulator breaks and to narrow down the responsible feature by comparison against a matched control.

Stochastic gene expression and its approximations. Two-state (telegraph) transcription is a workhorse model whose stationary distribution is the Poisson-Beta (9; 10), and its kinetics are routinely inferred from single-cell data (16; 17). The exact chemical master equation is solvable but costly (4), so inference commonly relies on cheaper approximations, the chemical Langevin equation (5) and the linear noise approximation, whose accuracy is known to deteriorate at low copy number and high burstiness (18; 15). Prior work adopts these approximations as a computational convenience (21); we quantify how that choice biases *neural* inference, show that in this model the failure tracks the Gaussian surrogates rather than cheap surrogates in general, and turn the closed-form Fano factor into an operational diagnostic for when the approximation is safe.

3 Setup

Biological background. The data are single-cell snapshots: for one gene, an experiment counts the integer number of mRNA molecules in each of n_{cells} cells, so each observation is an empirical distribution of counts. The exact generative model is the chemical master equation (CME), the forward equation of a continuous-time Markov jump process over molecule counts; it is exact but usually expensive to solve or simulate. The CLE and the LNA replace this discrete process with continuous Gaussian dynamics, which is cheaper but wrong when counts are small. Transcription is *bursty* when a gene switches on rarely and produces many molecules each time, which yields many cells with zero molecules (zero-inflation) and a heavy tail. We summarize burstiness by the Fano factor, the variance-to-mean ratio of the counts, which equals one for a Poisson distribution and grows with burstiness. Training on the CLE or LNA and querying on CME data is therefore a train/test mismatch in the likelihood.

Telegraph model and exact CME. A single gene switches between an inactive and an active state and transcribes mRNA only while active. We measure time in units of the mRNA lifetime (degradation rate $\gamma = 1$), so the free parameters are $\theta = (k_{\text{on}}, k_{\text{off}}, k_{\text{syn}})$: the switch-on, switch-off, and transcription rates. The stationary mRNA distribution takes the exact Poisson-Beta form (9; 10)

$$x \sim \text{Beta}(k_{\text{on}}, k_{\text{off}}), \qquad m \mid x \sim \text{Poisson}(k_{\text{syn}}\, x), \tag{1}$$

with closed-form mean and Fano factor

$$\mathbb{E}[m] = k_{\text{syn}} \frac{k_{\text{on}}}{k_{\text{on}} + k_{\text{off}}}, \qquad \text{Fano}[m] = 1 + \frac{k_{\text{syn}}\, k_{\text{off}}}{(k_{\text{on}} + k_{\text{off}})(k_{\text{on}} + k_{\text{off}} + 1)}. \tag{2}$$

Equation (1) gives our high-fidelity, ground-truth generator, and Equation (2) gives the analytic handle that the diagnostic later uses. We parametrize burstiness by the burst frequency $f = k_{\text{on}}$ and the mean burst size $b = k_{\text{syn}}/k_{\text{off}}$; low f with high b produces bursty, bimodal, zero-inflated expression and pushes the Fano factor well above one.

The cheap, misspecified simulator (CLE). The chemical Langevin equation treats the gene state $g \in [0, 1]$ and the mRNA count m as continuous variables driven by Gaussian reaction noise:

$$\mathrm{d}g = (k_{\text{on}}(1 - g) - k_{\text{off}} g)\,\mathrm{d}t + \sqrt{k_{\text{on}}(1 - g)}\,\mathrm{d}W_1 - \sqrt{k_{\text{off}} g}\,\mathrm{d}W_2, \tag{3}$$

$$\mathrm{d}m = (k_{\text{syn}} g - m)\,\mathrm{d}t + \sqrt{k_{\text{syn}} g}\,\mathrm{d}W_3 - \sqrt{m}\,\mathrm{d}W_4. \tag{4}$$

We integrate these equations by Euler-Maruyama with reflecting boundaries that clamp g to $[0, 1]$ and m to ≥ 0. The Gaussian, continuum approximation, applied to both the mRNA count and the single-copy gene state, creates the misspecification: it cannot represent the discrete, zero-inflated statistics of bursts, and the boundary leakage that forces the clamping is itself a symptom.

A second Gaussian surrogate (LNA). To test whether the failure is specific to the CLE or generic to Gaussian surrogates, we add the linear noise approximation (LNA), which models mRNA as a single Gaussian with mean and variance obtained from the linearized system. The LNA is constructed to match the exact CME mean and variance, but enforcing non-negative counts (we round each draw and clip it at zero) distorts them in the bursty regime, where a Gaussian centered near zero spills below it; it also cannot place the roughly 90% of mass at zero that a bursty gene produces. It is a cheaper and structurally different approximation than the CLE, and it lets us ask whether a surrogate aimed at the moments can still be trusted.

A non-Gaussian control (NB). To test whether the failure comes from the Gaussian family or merely from carrying the wrong moments, we add a negative binomial (NB) surrogate that uses the *same* analytic mean and Fano factor as the LNA but wraps them in a count distribution rather than a Gaussian. Given θ we compute the exact mean μ and Fano F from (2) and sample NB with success probability $1/F$ and dispersion $\mu/(F-1)$, which matches both moments. Unlike the LNA, the NB is non-negative and integer-valued by construction, so it needs no clamping and can represent overdispersion and zero-inflation; indeed the telegraph marginal approaches an NB in the bursty limit. The NB is thus a matched control that carries the same moments as the LNA in a different distributional family; it compares the two families directly, although, as a count distribution, it also avoids the clamping that the LNA needs.

Neural posterior estimation. Given a simulator $p_{\mathrm{sim}}(x \mid \theta)$, neural posterior estimation (NPE) trains a conditional density estimator $q_\phi(\theta \mid x)$ by maximizing $\mathbb{E}_{\theta\sim p(\theta),\, x\sim p_{\mathrm{sim}}(\cdot\mid\theta)}[\log q_\phi(\theta \mid x)]$, and thereby recovers the simulator-implied posterior (2; 3). We use a masked autoregressive flow (22) through the `sbi` toolkit (11). Misspecification enters when the training simulator differs from the data generator: an estimator that we train on the CLE targets $p_{\mathrm{CLE}}(\theta \mid x)$, yet we query it on data $x \sim p_{\mathrm{CME}}(\cdot \mid \theta)$.

4 Experimental design

We place a broad uniform prior over $\log_{10}$ rates ($\log_{10} k_{\mathrm{on}} \in [-2, 1.3]$, $\log_{10} k_{\mathrm{off}} \in [-1, 1.3]$, $\log_{10} k_{\mathrm{syn}} \in [0, 2.3]$). Each observation x collects a snapshot of $n_{\mathrm{cells}} = 500$ mRNA counts, which we reduce to six summary statistics: the log mean, log variance, Fano factor, fraction of zero-count cells, and the log median and log 90th percentile (all logs are $\log(1+\cdot)$). These features deliberately expose what a Gaussian approximation misrepresents, namely the zero fraction and the super-Poissonian tail. We train four estimators, one each on $8{,}000$ CME, CLE, LNA, and NB simulations, and we evaluate all of them on the same held-out set of 400 snapshots drawn from the exact CME, which isolates the effect of the training simulator. As an ablation, we also rerun the whole pipeline with only the mean and variance as features, to test whether the failure depends on this feature choice. Each estimator is a masked autoregressive flow (5 transforms, 50 hidden features) trained with the `sbi` defaults: Adam at learning rate 5×10^{-4}, batch size 200, a 10% validation split, and early stopping after 20 epochs without validation improvement.

We report two quantities as a function of burstiness, which we bin by the *empirical* Fano factor computed from each snapshot, the same observable quantity the routing rule below uses: (i) posterior-mean error, the absolute error of the posterior mean against the true θ in $\log_{10}$ units; and (ii) calibration, through the empirical coverage of 90% credible intervals, expected-coverage curves across all credibility levels, and simulation-based calibration ranks (12; 13). We draw bootstrap 95% intervals over test points within each bin (equal-count sextiles of the shared test set), and we confirm that the empirical Fano factor tracks the analytic Fano factor of (2) with correlation of about 0.9 at our 500-cell snapshots and rising toward one as the snapshot grows (Section 5.6), so binning by the observable proxy faithfully stratifies by true burstiness.

5 Results

5.1 The cheap simulator misspecifies the model directionally, and only where it matters

Figure 1 sweeps the burst frequency k_{on} at fixed $k_{\mathrm{off}} = 1$ and $k_{\mathrm{syn}} = 40$ and compares exact CME statistics against CLE statistics. In the fast-switching, non-bursty regime the two track closely (their means differ by about 13 percent at the fastest switching). As burstiness rises, the CLE fails in one consistent direction: it overestimates the mean, suppresses the Fano factor, and erases the zero-inflation. At the bursty end the true fraction of zero-mRNA cells reaches 0.90 while the CLE reports ≈ 0.04; the Fano factor reaches 20.3 for the CME against ≈ 4.6 for the CLE; and the mean sits at ≈ 1.0 for the CME against ≈ 5.7 for the CLE. This discrepancy is not diffuse noise but a structured, one-sided distortion that concentrates in the biologically interesting regime.

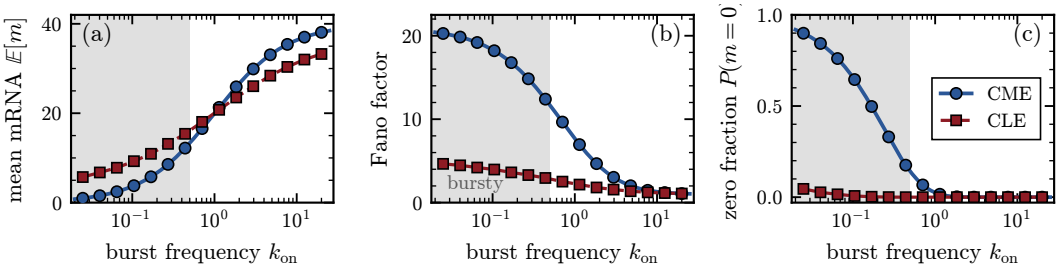


Figure 1: The CLE misspecifies the model directionally and only in the bursty regime. Across a burst-frequency sweep ($k_{\mathrm{off}} = 1$, $k_{\mathrm{syn}} = 40$), the cheap CLE simulator (red, dashed) matches the exact CME (blue) when switching is fast, but as burstiness grows it overestimates the mean (a), suppresses the Fano factor (b), and erases the zero-inflation (c) that defines bursty transcription. The shaded band marks the bursty regime. CME curves are exact; each CLE point simulates 20,000 cells.

5.2 Two Gaussian surrogates, two failure signatures

We first ask how the training simulator shapes inference on exact CME data. Figure 2 bins the test snapshots by empirical Fano factor and shows, for each estimator, the posterior-mean error (a) and the 90% coverage of k_{on} (b). Both Gaussian surrogates lose calibration relative to the CME-trained control, but they fail with different signatures. The CLE is catastrophic: its 90% credible intervals for k_{on} cover the truth only about 21% of the time overall (Table 1), below nominal in every bin and falling from 0.52 in the least bursty bin to 0.00 in the most bursty one, because its dynamics and boundary clamping distort the count distribution across the range. The LNA, built to match the first two moments, is intermediate (k_{on} coverage 0.55 overall) and its failure has an onset: its coverage stays between 0.63 and 0.74 in the four least bursty bins (empirical Fano below about 4.5) and collapses to 0.37 and 0.24 in the two most bursty ones, because enforcing non-negativity distorts its moments and it cannot reproduce the zero-inflation precisely where bursts dominate. Throughout, the control holds near the nominal 90%, with its largest error on near-Poisson snapshots, which say little about the switching rates. Averaged over the prior (Table 1), the CLE mis-estimates k_{on} by 1.35 $\log_{10}$ units, more than an order of magnitude, and the LNA by 0.71, against 0.17 for the control. The damage concentrates in k_{on} and k_{syn} and affects k_{off} much less, exactly the parameters that the zero fraction and tail encode and that a Gaussian erases. The reliability curves of Figure 3 show the same pattern at every credibility level. A further symptom reinforces the picture: queried on exact CME data, a misspecified flow places much of its posterior mass outside the prior support, so standard rejection sampling accepts essentially no draws and we sample every flow, including the control, directly. This leakage is a symptom rather than the cause: when we reparametrize the flow so that it cannot leave the prior box, the leaked mass vanishes for all four estimators, yet the surrogates stay miscalibrated (k_{on} coverage 0.60 for the LNA and 0.21 for the CLE, in a separate run with 2,000 test points), so the conclusions do not depend on how the posterior handles the prior support.

To separate non-Gaussian shape from our choice of summary statistics, we repeat the experiment with only the mean and variance as features. The well-specified CME control stays calibrated (k_{on} coverage 0.92), which rules out the feature set as the source of failure, but neither surrogate is rescued. The CLE stays at 0.23 and the LNA at 0.46, because enforcing non-negative counts distorts even their moments where bursts dominate, so a non-negative Gaussian surrogate for counts cannot be cheaply repaired by moment-matching. The ordering is also stable across randomness. Over five independent runs that resample the training data, re-initialize the network, and redraw the test set, the k_{on} 90% coverage is 0.91 ± 0.02 (CME), 0.57 ± 0.08 (LNA), and 0.19 ± 0.01 (CLE), so the separation between the control and both surrogates holds in every run.

5.3 The distributional family, not cheapness or the moments

If a Gaussian surrogate fails because it cannot represent a bursty count distribution, then a *cheap* surrogate that carries the same moment information but the right distributional family should succeed. The NB control tests exactly this (Section 3). Against exact CME data it is calibrated across the whole burstiness range, with k_{on} coverage 0.91 matching the CME control (Table 1, Figures 2 and 3), even though the LNA built on the identical moments sits at 0.55. The match is close rather than

Table 1: Prior-averaged inference quality against exact CME data. The two Gaussian surrogates degrade relative to the CME-trained control, ordered by how much they distort the bursty count distribution (LNA intermediate, CLE worst), while the NB, a non-Gaussian surrogate carrying the *same* analytic moments as the LNA, nearly matches the control (with a small residual gap on k_{syn}).

	posterior-mean abs. error ($\log_{10}$)			90% CI coverage		
Training simulator	k_{on}	k_{off}	k_{syn}	k_{on}	k_{off}	k_{syn}
CME (control)	0.17	0.35	0.14	0.89	0.89	0.90
NB (non-Gaussian control)	0.24	0.52	0.22	0.91	0.91	0.84
LNA (misspecified)	0.71	0.51	0.45	0.55	0.81	0.55
CLE (misspecified)	1.35	0.53	0.36	0.21	0.77	0.29

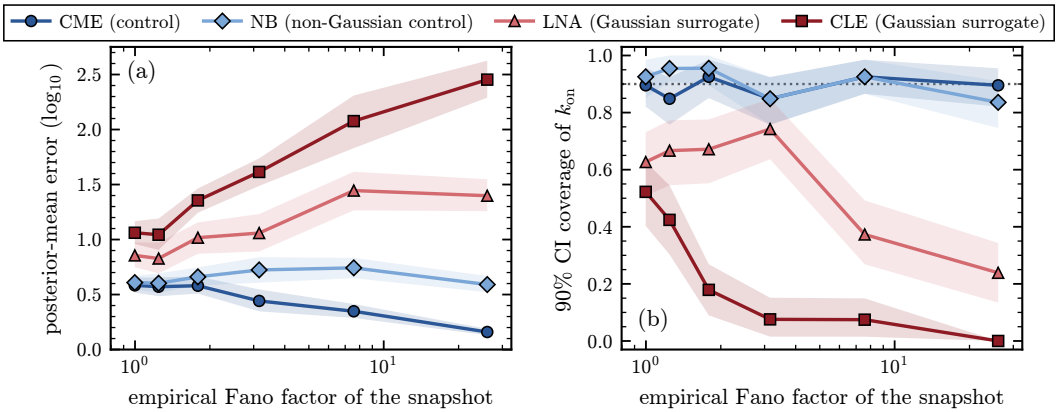


Figure 2: Inference degrades where the simulator misspecifies the model. Posterior-mean error, as the Euclidean norm over the three parameters (a), and 90% credible-interval coverage for k_{on} (b) as a function of the empirical Fano factor of each 500-cell snapshot, an observable quantity, binned in equal-count sextiles of one test set shared by all estimators, for estimators trained on the CLE (misspecified, dark red), the LNA (misspecified, light red), the NB (non-Gaussian control, light blue), and the CME (control, dark blue), all evaluated on exact CME data. Shaded bands are bootstrap 95% intervals over test points. The Gaussian surrogates fail: the CLE is miscalibrated in every bin and collapses toward zero coverage as burstiness grows, and the LNA holds up to an empirical Fano of about 4.5 and then collapses, while the NB, carrying the same moments as the LNA, tracks the CME control near the nominal 0.9 (dotted).

exact, since the telegraph marginal is only approximately negative binomial, so the NB keeps a small residual gap on k_{syn}; it is nonetheless far better calibrated than either Gaussian surrogate on every parameter. Same moments, same cost, opposite outcome: the failure is not caused by approximating the simulator cheaply, nor by the moments the surrogate carries, but by the distributional family it imposes on discrete, zero-inflated counts. The comparison contrasts two implemented families rather than isolating Gaussianity as a single causal factor, since the NB also needs no clamping (Section 6). It also shows what a trustworthy surrogate needs, a count family with the right dispersion, although here the NB borrows its parameters from the closed form.

5.4 A closed-form diagnostic predicts the breakdown, and a routing rule follows

Because the failure tracks burstiness for the moment-matched surrogate, and the analytic Fano factor (2) fixes burstiness a priori, we predict where even a good Gaussian surrogate turns unsafe without running any inference. The *analytic* form of the diagnostic, available only in a sandbox like this one, explains the mechanism by predicting the breakdown directly from θ; the *empirical* form reads the Fano factor off the snapshot, needs no closed form or knowledge of θ, and is the part that deploys elsewhere. The two agree closely (correlation about 0.9 at our snapshot size), so the mechanism transfers through the observable proxy. Empirically the LNA's coverage holds up to an empirical

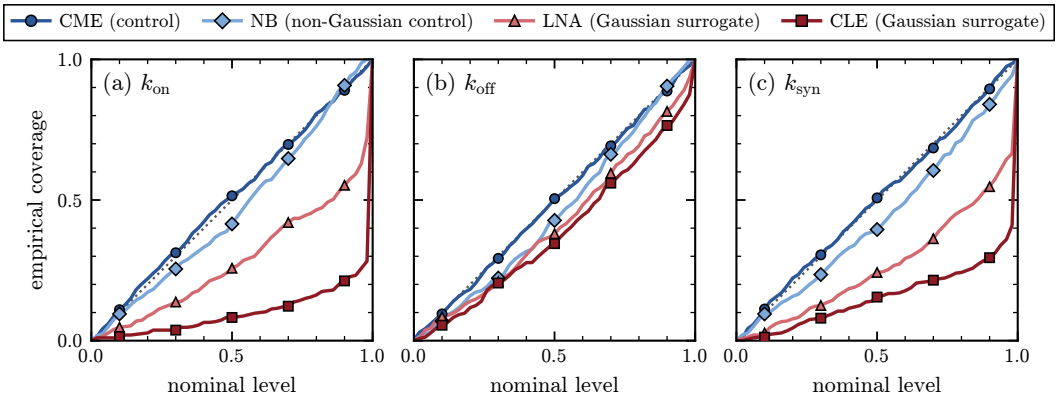


Figure 3: Reliability (expected-coverage) curves on exact CME data: empirical coverage of central credible intervals against their nominal level, per parameter (a to c) and training simulator. A calibrated estimator lies on the diagonal and an overconfident one falls below it. The CME control tracks the diagonal and the NB nearly so, with a small deficit on k_{syn}, whereas the Gaussian surrogates fall well below it on k_{on} and k_{syn}.

Fano of about 4.5 and collapses beyond it (Figure 2), while the CLE is unsafe across the whole range, so the diagnostic is informative for the moment-matched surrogate and moot for the CLE; we validate it as a per-observation routing signal in Section 5.5. We use $\mathrm{Fano}^{\star} \approx 4.5$ as a tunable operating point rather than a hard threshold. The two diagnostics then combine into a routing rule: (i) read the empirical Fano off each snapshot; (ii) choose the surrogate family by data type, not cost, since for these zero-inflated counts a negative binomial is safe where a Gaussian is not; and (iii) send queries with $\mathrm{Fano} > \mathrm{Fano}^{\star}$ to inference with the exact model, leaving the near-Poisson regime to the surrogate.

5.5 The routing rule, demonstrated

We validate the rule end to end. A population of inference queries spanning burstiness must each be served either by a cheap, misspecified estimator (LNA-trained) or, for a limited fraction f of queries, by the exact CME-trained estimator; we compare a Fano-guided policy, which routes the highest empirical-Fano queries to the exact estimator, against random routing. Serving every query with the cheap estimator yields 0.54 coverage on k_{on}, and serving every query exactly yields 0.91. Between these extremes the Fano-guided policy dominates random at every budget (Figure 4): at half the exact budget it already reaches about 0.78 coverage against about 0.72 for random, because the empirical Fano correctly picks out the high-burstiness queries where the surrogate fails. A query routed to the exact estimator stands for observation-specific exact inference, the expensive step in systems without a closed form; since exact simulation is cheap in our sandbox, the experiment tests whether the Fano ranks queries by surrogate failure, not how much computation the rule saves.

5.6 Robustness

As the snapshot size n_{cells} grows, the CME control stays calibrated while the CLE gets *worse* (its k_{on} coverage falls from 0.40 at $n_{\mathrm{cells}} = 100$ to 0.13 at 2000), the signature of genuine misspecification rather than finite-sample noise, since more data only sharpens an already wrong posterior. The observable diagnostic sharpens in step, with the correlation between empirical and analytic Fano rising from 0.76 at 100 cells to 0.998 at 2000. The ranking is stable across seeds (Section 5.2) and visible in Figure 3 and Appendix A.

6 Discussion and limitations

Three lessons generalize beyond this system. First, matching low-order moments is not a cheap fix, because non-negativity enforcement distorts the LNA's mean and variance in the bursty regime and its Gaussian shape cannot capture zero-inflation. Misspecification diagnostics should therefore probe

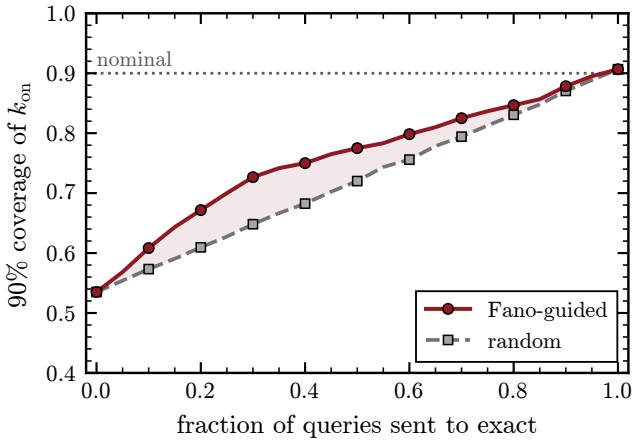


Figure 4: The routing rule in action. Overall 90% CI coverage for k_{on} against the exact-simulation budget, the fraction of queries served by the exact CME-trained estimator (the rest by the cheap LNA-trained one). Routing the highest empirical-Fano queries to the exact estimator (Fano-guided) dominates random allocation at every budget, recovering calibration with fewer queries sent to the exact model.

distributional shape rather than moment agreement alone; the Fano factor provides a convenient scalar proxy. Second, because NPE amortizes a single estimator over the prior, local surrogate errors can contaminate inference elsewhere, motivating our per-observation rule. Third, the NB control shows that, in this model, a cheap surrogate can recover calibration when its distributional family matches the data, suggesting that surrogate design should prioritize support and tail behavior over adding moments or defaulting to exact simulation.

Studying a single gene with exact ground truth lets us isolate simulator-induced failure and derive an analytic diagnostic, but it also limits our conclusions. We use the exact CME as a stand-in for real data and do not validate on experimental measurements, and we consider stationary snapshot summaries; experimental smFISH or scRNA-seq data, dynamic observations, and full-distribution inputs may change the observed thresholds. Our analytic claims are also restricted to the two-state model, while multi-state promoters, feedback, and coupled genes generally lack closed-form moments. The threshold $\mathrm{Fano}^\star$ is therefore system-, calibration-, and snapshot-size-dependent, an operating point rather than a universal constant. The NB control is imperfect because the telegraph marginal is only approximately negative binomial, and it takes its mean and Fano factor from the closed form, which other systems would have to estimate, for example from moment equations or a few exact simulations, at a cost we do not measure. Moreover, our attribution to the Gaussian family is partially confounded, because both Gaussian surrogates enforce non-negativity, which the NB does not need, and the CLE also relaxes the single-copy gene state to a continuous variable; an unclamped Gaussian control and a CLE with a discrete gene state would separate these effects. The weak identification of k_{off} from stationary snapshots also means its apparent robustness partly reflects near-prior posteriors rather than accurate recovery. Finally, exact simulation is cheap in this sandbox, so the routing rule illustrates rather than exercises a real computational trade-off. We characterize rather than correct misspecification.

7 Conclusion

Cheap approximate simulators are indispensable for scalable simulation-based inference, yet they silently import their own misspecification. Using the telegraph model, where the exact answer is known, we showed that Gaussian surrogates for stochastic gene expression corrupt neural inference in a directional, burstiness-dependent way, producing biased and over-confident posteriors exactly where the interesting biology lives. Two controls point to the distributional family imposed on discrete, zero-inflated counts, not cheapness or the moments a surrogate carries. A negative binomial matched on the same moments recovers calibration, which suggests a count-appropriate surrogate rather than more moments or the exact model everywhere. Where the surrogate is unavoidable, the Fano factor read from the snapshot predicts which queries it will mishandle, and Fano-guided routing recovers calibration with fewer exact queries than random routing. Its analytic form is confined to sandboxes, but its empirical form transfers, which we see as the route to multi-state, coupled, and real single-cell systems.

References

[1] Cranmer, K., Brehmer, J. & Louppe, G. (2020) The frontier of simulation-based inference. *Proceedings of the National Academy of Sciences* **117**(48):30055–30062.

[2] Papamakarios, G. & Murray, I. (2016) Fast ε-free inference of simulation models with Bayesian conditional density estimation. In *Advances in Neural Information Processing Systems 29*.

[3] Greenberg, D., Nonnenmacher, M. & Macke, J. (2019) Automatic posterior transformation for likelihood-free inference. In *International Conference on Machine Learning*, pp. 2404–2414.

[4] Munsky, B. & Khammash, M. (2006) The finite state projection algorithm for the solution of the chemical master equation. *Journal of Chemical Physics* **124**(4):044104.

[5] Gillespie, D.T. (2000) The chemical Langevin equation. *Journal of Chemical Physics* **113**(1):297–306.

[6] Cannon, P., Ward, D. & Schmon, S.M. (2022) Investigating the impact of model misspecification in neural simulation-based inference. *arXiv:2209.01845*.

[7] Ward, D., Cannon, P., Beaumont, M., Fasiolo, M. & Schmon, S.M. (2022) Robust neural posterior estimation and statistical model criticism. In *Advances in Neural Information Processing Systems 35*.

[8] Schmitt, M., Bürkner, P.-C., Köthe, U. & Radev, S.T. (2023) Detecting model misspecification in amortized Bayesian inference with neural networks. In *DAGM German Conference on Pattern Recognition*.

[9] Peccoud, J. & Ycart, B. (1995) Markovian modeling of gene-product synthesis. *Theoretical Population Biology* **48**(2):222–234.

[10] Raj, A., Peskin, C.S., Tranchina, D., Vargas, D.Y. & Tyagi, S. (2006) Stochastic mRNA synthesis in mammalian cells. *PLoS Biology* **4**(10):e309.

[11] Tejero-Cantero, A., Boelts, J., Deistler, M., Lueckmann, J.-M., Durkan, C., et al. (2020) `sbi`: A toolkit for simulation-based inference. *Journal of Open Source Software* **5**(52):2505.

[12] Talts, S., Betancourt, M., Simpson, D., Vehtari, A. & Gelman, A. (2018) Validating Bayesian inference algorithms with simulation-based calibration. *arXiv:1804.06788*.

[13] Hermans, J., Delaunoy, A., Rozet, F., Wehenkel, A., Begy, V. & Louppe, G. (2022) A trust crisis in simulation-based inference? Your posterior approximations can be unfaithful. *Transactions on Machine Learning Research*.

[14] Papamakarios, G., Nalisnick, E., Rezende, D.J., Mohamed, S. & Lakshminarayanan, B. (2021) Normalizing flows for probabilistic modeling and inference. *Journal of Machine Learning Research* **22**(57):1–64.

[15] Schnoerr, D., Sanguinetti, G. & Grima, R. (2017) Approximation and inference methods for stochastic biochemical kinetics: a tutorial review. *Journal of Physics A: Mathematical and Theoretical* **50**(9):093001.

[16] Munsky, B., Neuert, G. & van Oudenaarden, A. (2012) Using gene expression noise to understand gene regulation. *Science* **336**(6078):183–187.

[17] Larsson, A.J.M., Johnsson, P., Hagemann-Jensen, M., et al. (2019) Genomic encoding of transcriptional burst kinetics. *Nature* **565**(7738):251–254.

[18] Grima, R., Thomas, P. & Straube, A.V. (2011) How accurate are the nonlinear chemical Fokker-Planck and chemical Langevin equations? *Journal of Chemical Physics* **135**(8):084103.

[19] van Kampen, N.G. (2007) *Stochastic Processes in Physics and Chemistry*, 3rd ed. Amsterdam: North-Holland.

[20] Elf, J. & Ehrenberg, M. (2003) Fast evaluation of fluctuations in biochemical networks with the linear noise approximation. *Genome Research* **13**(11):2475–2484.

[21] Fröhlich, F., Thomas, P., Kazeroonian, A., Theis, F.J., Grima, R. & Hasenauer, J. (2016) Inference for stochastic chemical kinetics using moment equations and system size expansion. *PLoS Computational Biology* **12**(7):e1005030.

[22] Papamakarios, G., Pavlakou, T. & Murray, I. (2017) Masked autoregressive flow for density estimation. In *Advances in Neural Information Processing Systems 30*.

Contributions of LatinX authors

Both authors are LatinX researchers at Universidad de los Andes, Colombia. Rafael Velásquez, the presenting author, designed the study, developed the simulation and inference pipeline, ran all experiments, and wrote the manuscript. Pablo Arbeláez supervised the project and revised the manuscript.

NeurIPS Paper Checklist

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   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and introduction claim a directional, regime-localized misspecification of the CLE, a distinct pair of failure signatures for two Gaussian surrogates that isolates non-Gaussian shape as the cause, and a closed-form diagnostic that predicts the failure. Sections 5.1 and 5.2 support each claim with the corresponding experiment, and we explicitly frame the work as a controlled-sandbox characterization rather than a new inference method.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: The Discussion and limitations section states the scope conditions directly: the single-gene setting, the use of the exact CME as a stand-in for real data, the reliance on stationary snapshot statistics, the restriction to the two-state model where closed-form moments exist, and the fact that we characterize rather than correct the misspecification.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

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   Justification: The paper proves no new theorems. It uses established closed-form results for the telegraph model (the Poisson-Beta stationary distribution and its moments), with citations, and states the assumptions under which they hold (a single gene copy, degradation rate normalized to one, stationarity).

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   Justification: The paper specifies the exact model and the four simulators (CME, CLE, LNA, NB), the prior ranges, the summary statistics and the moments-only ablation, the number of simulations and cells, the density estimator with its architecture and training hyperparameters stated in the main text, and the evaluation protocol. The accompanying anonymized code reproduces every figure end to end from fixed random seeds.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results?

   Answer: [Yes]

   Justification: All data are synthetic and the provided code generates them. We include the anonymized simulation and inference code with a README that lists the exact commands, dependencies, and tested package versions.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details necessary to understand the results?

   Answer: [Yes]

   Justification: Section 4 gives the prior, the six summary statistics, the training-set sizes, the masked autoregressive flow through the `sbi` toolkit, and the early-stopping criterion; test data come from the exact CME, and evaluation uses posterior-mean bias, simulation-based calibration ranks, and 90% interval coverage.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [Yes]

   Justification: The money plot reports per-bin means with bootstrap 95% intervals over 400 test points, and the four estimators separate cleanly across bins. The effect sizes are large: the CLE's k_{on} bias exceeds an order of magnitude and its 90% coverage sits near 0.2 against a nominal 0.9. The ordering is stable across five random seeds, with per-run k_{on} coverage standard deviation at most 0.08.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources needed to reproduce the experiments?

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   Justification: All experiments run on a single CPU machine. Generating the CLE training set is the bottleneck at roughly six to eight minutes for $8{,}000$ simulations, cached once; each estimator trains in under two minutes. The README states these costs.

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    Question: Are the creators or original owners of assets used in the paper properly credited and are the license and terms of use explicitly mentioned and properly respected?

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    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

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Justification: Large language models are not a component of the core method. Any use was limited to writing and editing assistance, which does not require declaration.

A Simulation-based calibration ranks

Simulation-based calibration (12) checks a posterior estimator without reference posteriors. For each test snapshot we draw posterior samples, record the rank of the true parameter among them, and normalize it to $[0, 1]$. A calibrated estimator produces uniform ranks, so each histogram in Figure 5 should stay flat within the grey band. Departures have a direct reading. A spike at one edge means the truth falls systematically on one side of the posterior, a one-sided bias; spikes at both edges, a U-shape, mean the posterior is too narrow and therefore overconfident; and a bump in the middle means it is too wide. The histograms use the same 400 test snapshots as Table 1 and Figures 2 and 3, and the reliability curves of Figure 3 summarize the same ranks.

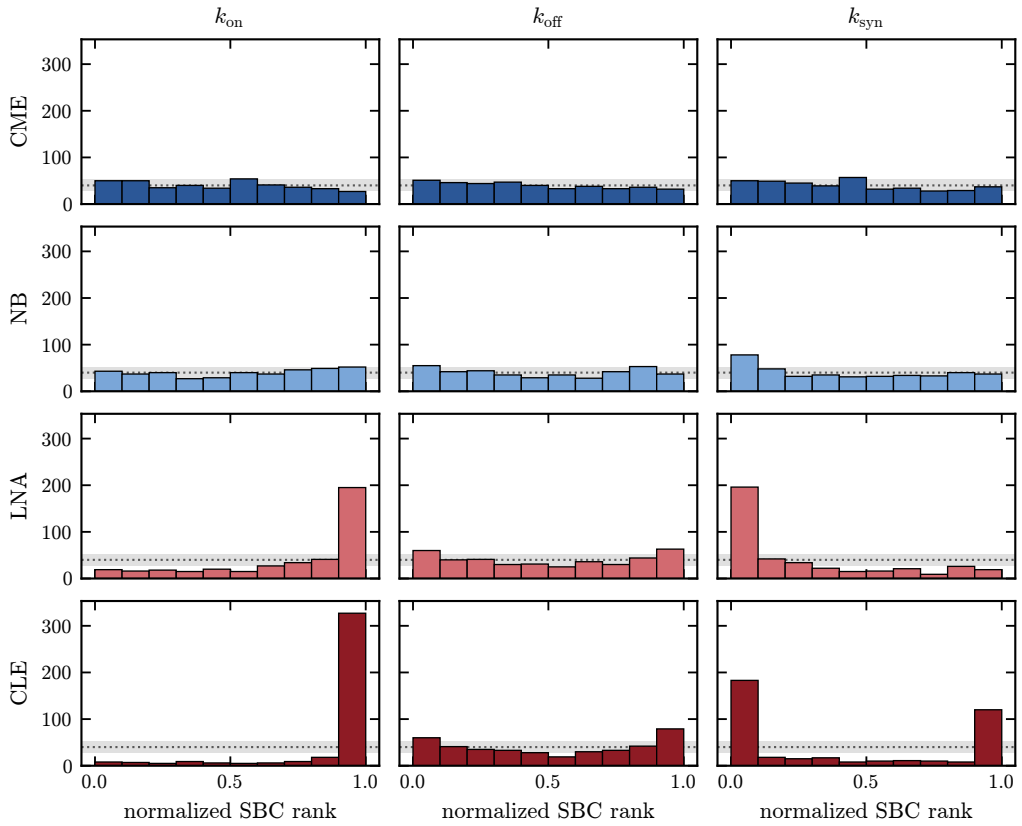


Figure 5: Simulation-based calibration ranks (rows: training simulator; columns: parameter). A calibrated estimator produces uniform ranks, flat within the grey band (two binomial standard deviations around the uniform count, dotted; about 95%). The CME control is flat throughout and the NB nearly so, apart from a mild k_{syn} edge, whereas the Gaussian surrogates deviate in an interpretable direction: edge spikes for k_{on} mark a one-sided bias, and the U-shape for the CLE on k_{syn} marks overconfident, too-narrow posteriors.